

Course Map and the Semester Project

DATA 443/643 · Advanced Concepts in Large Language Models

Week 1

American University

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Ideas and 16:9 layout from Stanford CS224N, CS231N, and MIT 6.S191. Original slides; those courses are not copied.

Lecture plan

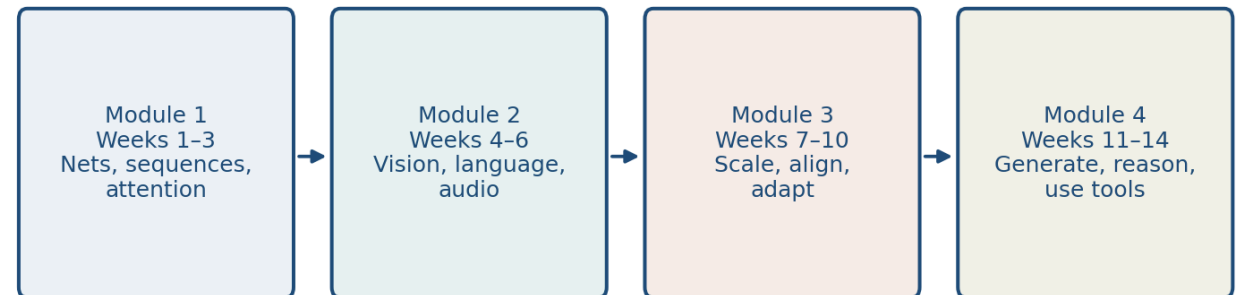
- 1 Four modules, taught bottom-up
- 2 The same object as a stack; Week 1 is the base
- 3 What those weights were trained to do
- 4 The weights file is 140 GB
- 5 The project names one layer

Key goal: one stack picture, one next-token loss, one measurable project sentence.

Four modules, one object

- Listed prereq: DATA 427/627. DATA 641 and 642 are not required.
- Overlap is intentional: neural nets and embeddings are taught here.
- The semester job is a measured project, not four electives.

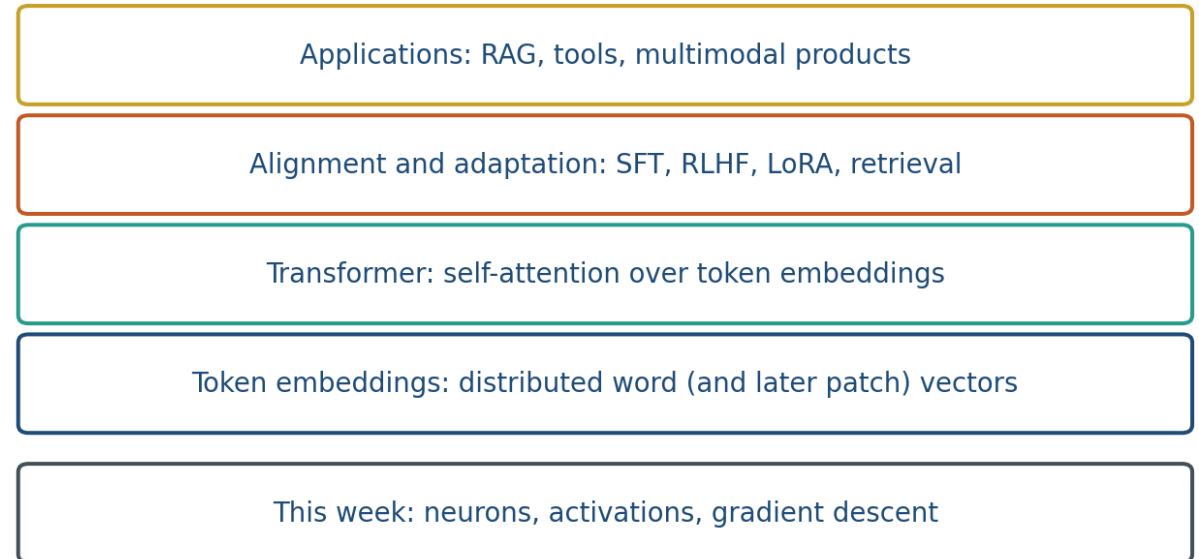
Where Week 1 sits: foundations before transformers



The same object as a stack

- Chat products sit at the top: decoding, tools, a UI.
- The weights still came from next-token training at the bottom.
- Name the layer you will change: data, prompt, adapter, retriever, or eval.

A language model is a stack. Week 1 is the bottom two layers.



What those weights were trained to do

$$P(w_1, \dots, w_T) = \prod_{t=1}^T P(w_t \mid w_{1:t-1})$$

Each factor is a softmax over the vocabulary.

Pretraining minimizes next-token NLL: minus the log of those probabilities.

If the model assigns 0.70, NLL is about 0.36. If 0.10, NLL is about 2.30.

Those numbers live in a 140 GB file

$$70 \times 10^9 \times 2 \text{ bytes} = 140 \text{ GB}$$

Llama 2 70B, each parameter stored as float16 (2 bytes).

Parameters file vs run file: running needs extra memory.

Karpathy: an LLM is weights plus a little code.

The project names one layer

A method

Use LoRA on news.

A question plus a metric

Does a LoRA adapter on local news reduce entity hallucination vs. the base model, measured by exact-match on a 100-item holdout?

Today's two-hour meeting

1.1 Course map

Stack, next-token NLL, 140 GB, project sentence. Short Karpathy clip.

1.2–1.4 Board notes

Neuron and XOR, then gradient descent, then skip-gram and bias.

Lab 1

Linear vs MLP on XOR, then the constructed 2-D embedding probe.

Key goal: one timed meeting, one stack picture, one project sentence.